

Example 9.2

Suppose we have a process $\{X_t\}$ with true form $MA(1)$ where

$$X_t = Z_t + \theta Z_{t-1} \quad (1)$$

Now, suppose we fit

$$X_t = Z'_t + \theta_0 Z'_{t-1} \quad (2)$$

where $\theta_0 \neq \theta$

「 Z'_t is not necessarily white noise」

Equate (1) and (2):

$$Z'_t + \theta_0 Z'_{t-1} = Z_t + \theta Z_{t-1}$$

$$(1 + \theta_0 B) Z'_t = (1 + \theta B) Z_t$$

If $\theta_0 \neq \theta$ then $\{Z'_t\}$ is an $ARMA(1,1)$ process

$$Z'_t = \frac{(1 + \theta B)}{1 + \theta_0 B} Z_t$$

Residuals $\{Z'_t\}$ are an $ARMA(1,1)$ process.